RESEARCH PAPER

Instructor Performance and Course Quality Evaluation on EduPro

A Data-Driven Study of Teacher Ratings, Experience, Course Quality and Transaction Activity

Prepared for

Unified Mentor Pvt. Ltd.

Submitted by

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Role

Business Analyst (Temporary Employment)

Internship period

25 September 2026 to 25 October 2026 (one month)

Report Information

This report documents the Business Analyst project completed during a one-month temporary employment with Unified Mentor Pvt. Ltd. The details of the internship and of the deliverables are given below.

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Organization	Unified Mentor Pvt. Ltd.
Role	Business Analyst (Temporary Employment)
Internship period	25 September 2026 to 25 October 2026
Duration	One month
Project title	Instructor Performance and Course Quality Evaluation on EduPro
Dataset	EduPro workbook, calendar year 2025: 60 instructors, 60 courses, 3,000 learners and 10,000 transactions
Tools used	Python (pandas, NumPy, SciPy, Matplotlib), Plotly, Streamlit
Live dashboard	https://edupro-quality-dashboard-kani-2026-3yzchnyaqxhaqhmwdgsucq.streamlit.app/

Abstract

Purpose. Online learning platforms depend on the quality of their instructors and of the courses those instructors deliver, yet many platforms judge both without a structured method. This study builds a data-driven framework to evaluate instructor performance and course quality on EduPro, an online education platform, and asks whether teaching experience, subject expertise and instructor rating are linked to course quality and to recorded transaction activity.

Dataset and methods. The analysis uses the EduPro workbook for calendar year 2025: 60 instructors, 60 courses, 3,000 learners and 10,000 transactions. The Teachers, Courses and Transactions sheets were integrated through TeacherID and CourseID and checked for missing values, duplicate identifiers and unmatched records. The study applies descriptive statistics, Pearson and Spearman correlation, simple linear regression, Kruskal-Wallis and Mann-Whitney group comparisons, and five KPIs, and the results are delivered through a Streamlit dashboard.

Main findings. The average teacher rating is 3.125 and the average course rating is 3.10. Twenty-one of 60 instructors (35.0%) are rated below 3.0, and all of them have six years of experience or less. Experience has a moderate positive association with teacher rating (Pearson $r = 0.598$, Spearman $\rho = 0.671$, $p < 0.001$), equal to about 0.12 rating points per year. Teacher rating, however, shows no relationship with the rating of the courses linked to the teacher ($r = 0.000$), and course quality varies far more between individual courses than between categories. Twenty of the 60 courses are rated below 2.5, and these courses account for 29.9% of revenue. Among 58 of the 60 instructors, transaction activity is unrelated to rating ($r = -0.107$). Two instructors hold 60.86% of all transactions through the same 454 learners, a pattern the data cannot explain.

Recommendations. EduPro should first validate the two-instructor pattern and close its data gaps (no instructor link on courses, no review counts, no dated ratings). It should then review its weakest courses one by one, support early-career instructors, adopt an evaluation scorecard that does not use transaction volume, and monitor the proposed KPIs regularly.

Keywords: instructor performance; course quality; teacher rating; online education analytics; exploratory data analysis; correlation analysis; transaction activity; key performance indicators; Streamlit.

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1. Introduction

1.1 Background of online education

Online education has moved from a supplement to a mainstream way of learning. A learner on an online platform cannot sit in a classroom and judge a teacher before enrolling. The decision rests on ratings, reviews and the reputation of the platform. Evidence reviewed by the U.S. Department of Education suggests that online learning can match or exceed face-to-face instruction when it is well designed (Means et al., 2010), which makes the design and delivery of each course a central concern for any platform.

1.2 Importance of instructor effectiveness

Instructors shape what learners experience in a course: how clearly ideas are explained, how learners are supported and how the material is paced. When instructor quality is uneven, learners meet very different experiences on the same platform, and weaker experiences lead to low ratings, negative reviews and lost trust. Knowing which instructors deliver consistently, and which need support, is therefore a basic management need.

1.3 Importance of course quality evaluation

A course is more than its instructor. Content, structure, level and length all affect the learner. For this reason, course quality has to be evaluated on its own, and not inferred from instructor quality. This study treats the two as separate measures and tests how closely they are related.

1.4 Need for data-driven decision-making

Without structured analysis, instructor evaluation remains subjective and fragmented. Decisions about hiring, training and course investment then depend on impressions. A data-driven framework replaces impressions with measurable indicators, makes weak spots visible early, and allows progress to be tracked over time. This paper provides such a framework for EduPro.

1.5 About the project

The research was carried out by Kanimozhi Ravi (UNID: UMID210926110647) as a Business Analyst project for Unified Mentor Pvt. Ltd., during a one-month temporary Business Analyst role from 25 September 2026 to 25 October 2026. It covers the EduPro workbook for 2025 and focuses on teaching effectiveness and course quality. It does not study learner behaviour, learning outcomes or review text, because these are not in the data.

2. Problem Statement

EduPro needs to understand the following, and currently has no structured way to do so:

- Which instructors consistently deliver high-quality courses?
- Does teaching experience relate to higher instructor and course ratings?
- Which course categories and levels perform best?
- Does instructor expertise relate to course quality?
- Are higher-rated instructors associated with higher recorded transaction activity?
- How evenly is teaching performance distributed across the platform?

Without answers, strong instructors are not reliably identified, weaker ones are not supported, and the platform has no evidence to guide training or course investment.

3. Aim and Objectives

3.1 Aim

To build and apply a data-driven framework that evaluates instructor performance and course quality on EduPro and identifies where the platform should act to improve teaching outcomes.

3.2 Objectives

- To integrate the instructor, course and transaction data into one validated dataset.
- To describe the distribution of instructor ratings, experience, age and expertise.
- To measure how experience relates to teacher rating and to course rating.
- To compare course quality across categories, levels and course types.
- To examine whether instructor rating is associated with recorded transaction activity.
- To define five KPIs with transparent formulas and calculate them from the data.
- To identify weak instructors, weak courses and data gaps, and to recommend actions.
- To deliver the findings through an interactive Streamlit dashboard.

4. Research Questions

- RQ1. Instructor rating distribution: how are instructor ratings distributed across the platform?
- RQ2. Experience versus instructor rating: do instructors with more experience receive higher ratings?

- RQ3. Instructor rating versus course rating: is a higher teacher rating linked to higher-rated courses?
- RQ4. Expertise-wise performance: which expertise areas show the strongest instructor and course ratings?
- RQ5. Course category and level: which categories and levels perform best and worst?
- RQ6. Transaction activity versus instructor rating: do higher-rated instructors record more transactions?

5. Dataset Description

5.1 Sheets and fields

The EduPro workbook has four sheets: Teachers (60 rows), Courses (60 rows), Transactions (10,000 rows) and Users (3,000 rows). The study uses the Teachers, Courses and Transactions sheets as specified in the project brief, and uses additional fields from the same workbook (such as CourseType, CoursePrice, TransactionDate and Amount) for revenue and trend analysis. The Users sheet is used only to count learners.

Table 1: Data dictionary

Sheet	Field	Type	Key	Description
Teachers	TeacherID	Text	PK	Unique instructor identifier (TC00001 to TC00060)
Teachers	TeacherName	Text		Instructor name
Teachers	Age	Integer		Instructor age in years
Teachers	Gender	Text		Instructor gender
Teachers	Expertise	Text		Subject area of the instructor (12 values)
Teachers	YearsOfExperience	Integer		Years of teaching experience
Teachers	TeacherRating	Decimal		Instructor rating (observed 1.05 to 4.97)
Courses	CourseID	Text	PK	Unique course identifier (CR00001 to CR00060)
Courses	CourseName	Text		Course title (two titles repeat under different IDs)
Courses	CourseCategory	Text		Subject category (12 values, 5 courses each)
Courses	CourseType	Text		Free or Paid
Courses	CourseLevel	Text		Beginner, Intermediate or Advanced
Courses	CoursePrice	Decimal		Course price (currency not stated)
Courses	CourseDuration	Decimal		Course duration (unit not stated)
Courses	CourseRating	Decimal		Course rating (observed 1.13 to 4.94)

Sheet	Field	Type	Key	Description
Transactions	TransactionID	Text	PK	Unique transaction identifier
Transactions	UserID	Text	FK	Learner who made the transaction (links to Users)
Transactions	CourseID	Text	FK	Course in the transaction (links to Courses)
Transactions	TeacherID	Text	FK	Instructor recorded on the transaction (links to Teachers)
Transactions	TransactionDate	Date		Date of the transaction (2025-01-01 to 2025-12-30)
Transactions	Amount	Decimal		Amount recorded (zero for free courses)
Transactions	PaymentMethod	Text		Bank Transfer, PayPal or Credit Card
Users	UserID	Text	PK	Unique learner identifier
Users	UserName, Age, Gender, Email	Mixed		Learner attributes (not used in the analysis)

5.2 Primary keys and relationships

TeacherID is the primary key of Teachers, CourseID is the primary key of Courses and TransactionID is the primary key of Transactions. The Transactions sheet holds TeacherID, CourseID and UserID as foreign keys. Each transaction therefore points to exactly one teacher and one course, which gives two many-to-one relationships (Transactions to Teachers and Transactions to Courses). Teachers and courses are related only through Transactions, as a many-to-many relationship.

The Courses sheet has no TeacherID. There is no official record of which instructor owns which course. The data contains 887 different teacher-course pairs, each course appears with 7 to 30 teachers, and each teacher appears with 7 to 55 courses. For this reason, no teacher-to-course ownership is assumed in this paper. Wherever a teacher is compared with courses, the comparison refers to the courses that appear next to that teacher in transactions.

6. Research Methodology

6.1 Data loading and cleaning

The workbook was loaded with pandas. Dates were parsed as dates and rating, price, duration and amount fields as numbers. Course titles "Deep Learning" and "Natural Language Processing" each appear twice under different CourseIDs and categories, and they were kept as separate courses because every result is grouped by CourseID. One course (CR00028, Deep Learning in Machine Learning) is priced at 0.78 while other paid courses are priced between

37.89 and 490.90. The price is consistent across all its transactions, so it was left unchanged and flagged for confirmation.

6.2 Missing values, duplicates and unmatched records

Each sheet was checked for missing values, duplicate identifiers and fully duplicated rows. Transactions were then checked against the other sheets for course, teacher and user identifiers with no match. The results are given in Table 2. No record needed to be imputed or removed.

6.3 Data integration

The Transactions sheet was joined to Courses on CourseID and to Teachers on TeacherID. Both joins were many-to-one on unique keys, and the script confirms that the row count stays at 10,000 after the joins. Teacher-level tables (rating, experience, transactions, learners) and course-level tables were built from the joined data. Learner counts are never added across teachers or courses, because one learner can appear in many.

6.4 Exploratory data analysis and descriptive statistics

The distribution of each variable was examined with histograms, bands and quartile statistics, and top and bottom performers were listed. Charts have titles, labelled axes and a short interpretation.

6.5 Correlation analysis and group comparisons

Pearson correlation measures linear association and Spearman rank correlation measures monotonic association, and both are reported because the sample is small and some values are extreme. A simple linear regression of teacher rating on experience gives the slope used as the Experience Impact Score. Differences between groups (categories, levels, expertise areas, free versus paid) were tested with the Kruskal-Wallis and Mann-Whitney tests. Statistical significance is reported only where a test was performed.

6.6 KPI calculation

Five KPIs were defined with transparent formulas (Section 9) and calculated from the data. Where transactions do not establish actual enrolment counts, the related KPI is labelled a transaction-activity proxy.

6.7 Streamlit dashboard development

A Streamlit application was built with pandas and Plotly to let users filter and explore the same data (Section 13). The analysis script that reproduces every number and chart in this paper is provided with the paper.

7. Exploratory Data Analysis

7.1 Data-quality summary

Before any analysis, the workbook was checked for completeness and consistency. Table 2 summarises the checks.

Table 2: Data-quality checks

Check	Result	Comment
Missing values (all four sheets)	0	No imputation needed
Duplicate IDs (user, teacher, course, transaction)	0	All keys are unique
Fully duplicated rows	0	
Transactions with no matching course	0	CourseID join is complete
Transactions with no matching teacher	0	TeacherID join is complete
Transactions with no matching user	0	All 3,000 users appear in Transactions
Teachers with no transaction	0	Every teacher appears at least 23 times
Courses with no transaction	0	Every course appears 140 to 196 times
Repeated course titles	2 titles (4 rows)	Different CourseIDs; kept separate
Rating range, teachers / courses	1.05-4.97 / 1.13-4.94	Scale assumed 1 to 5; not stated in the data
Transaction dates	2025-01-01 to 2025-12-30	One calendar year only

The data is complete and internally consistent. The data-quality concerns are structural and not about errors: there is no TeacherID on Courses, no review counts behind each rating, no dates on ratings, no stated currency or duration unit, and an unexplained pattern in two instructors' transactions (Section 10).

7.2 Instructor rating distribution

Teacher ratings run from 1.05 to 4.97, with a mean of 3.125 and a median of 3.275. The median is above the mean because of a longer tail of low ratings (skewness -0.485). Table 3 and Figure 1 show how the 60 instructors fall into rating bands.

Table 3: Teachers and courses by rating band

Rating band	Teachers	Share	Courses	Share
1.0 to 1.99	9	15.0%	13	21.7%
2.0 to 2.99	12	20.0%	16	26.7%
3.0 to 3.99	28	46.7%	15	25.0%
4.0 to 5.0	11	18.3%	16	26.7%

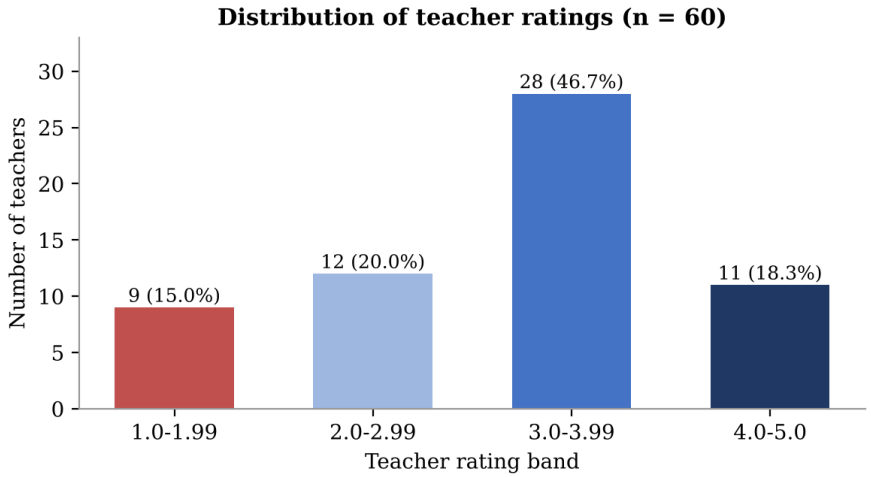


Figure 1: Distribution of teacher ratings by band

Interpretation. Only 11 of 60 instructors (18.3%) reach a rating of 4.0, while 21 (35.0%) fall below 3.0 and nine fall below 2.0. Teaching performance is therefore not evenly distributed: a third of the staff sits in the lower bands. Course ratings are spread even more widely than teacher ratings (standard deviation 1.17 against 0.95).

7.3 Instructor experience distribution

Experience ranges from 1 to 24 years with a mean of 6.28 and a median of 6 years. The distribution is skewed to the right: 36 of the 60 instructors have six years or less, and only three have more than 14 years. Instructor age ranges from 27 to 50 years (mean 38.5).

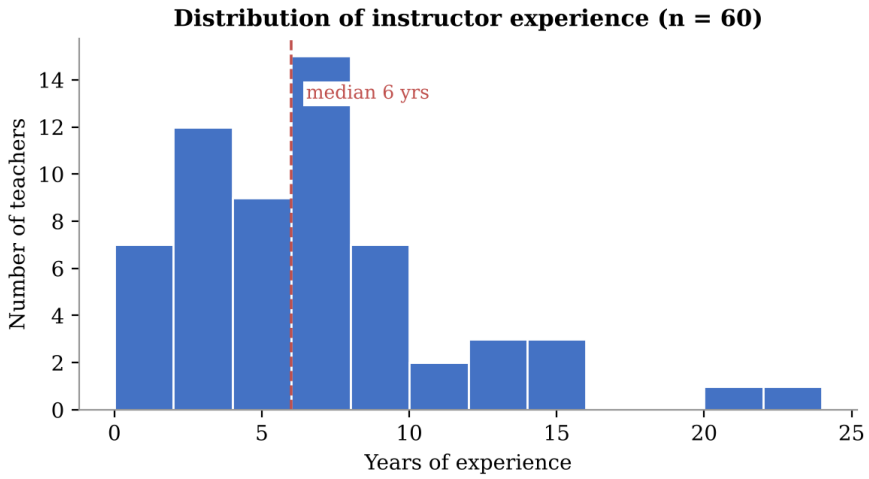


Figure 2: Distribution of instructor experience

Interpretation. The platform relies mostly on early-career instructors. Any conclusion about very experienced teachers rests on a small group, which is why results are also checked without the longest-serving instructors.

7.4 Experience versus teacher rating

The scatter plot below shows a clear upward pattern. The Pearson correlation is 0.598 and the Spearman correlation is 0.671 (both $p < 0.001$). The fitted line rises by 0.120 rating points for each additional year of experience.

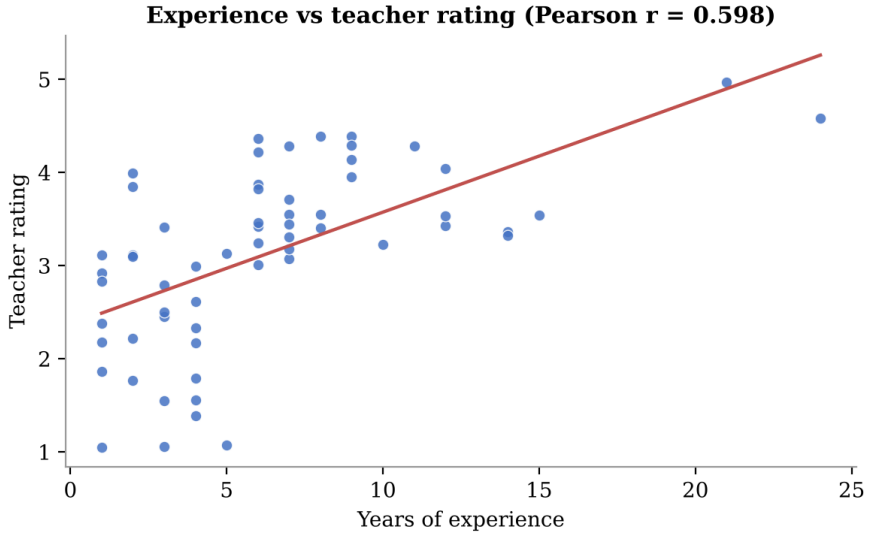


Figure 3: Years of experience against teacher rating, with linear fit

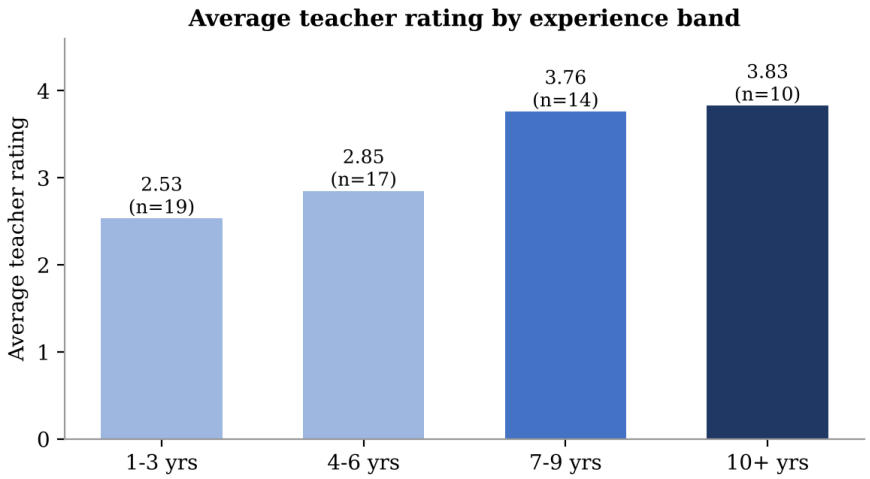


Figure 4: Average teacher rating by experience band

Interpretation. Ratings rise steadily up to about seven years and then level off: instructors with one to three years average 2.53, four to six years 2.85, seven to nine years 3.76 and ten or more years 3.83. This flattening suggests diminishing returns after about seven years, although it is a pattern in group averages and not a proven threshold. Experience is not destiny: two instructors with only two years of experience are rated 3.99 and 3.85.

7.5 Experience and teacher rating versus course rating

The Courses sheet has no TeacherID, so a course rating cannot be assigned to one instructor. To approximate it, the average rating of the courses that appear next to each teacher in transactions was calculated. Because most teachers appear with many different courses, this is a blunt measure, and the result must be read lightly.

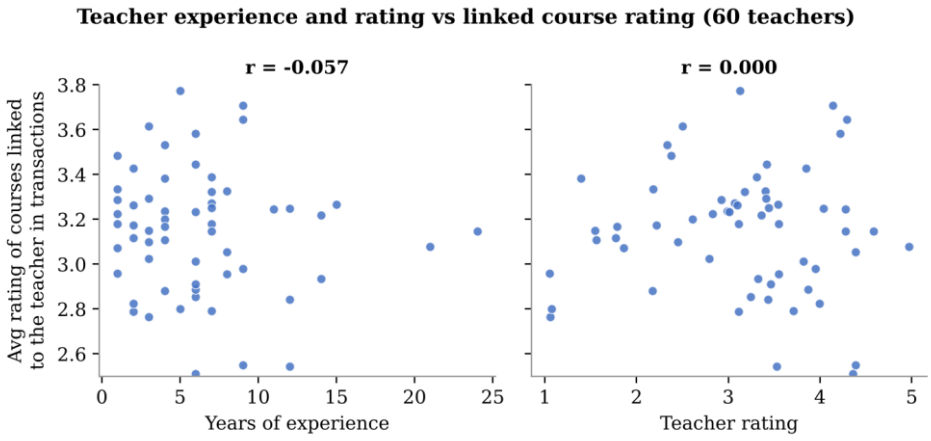


Figure 5: Experience and teacher rating against the rating of linked courses

Interpretation. Neither experience (r = -0.057, p = 0.666) nor teacher rating (r = 0.000, p = 0.997) is related to the rating of the linked courses. In other words, a highly rated instructor is

not tied to higher-rated courses in this data. This points towards course content and design, and not only the instructor, as the source of course quality. It is not proof, because of the weak link between teachers and courses.

7.6 Course rating by category and level

Course ratings average 3.10 and range from 1.13 to 4.94. Each of the 12 categories has exactly five courses, so one very high or very low course can move a category average noticeably.

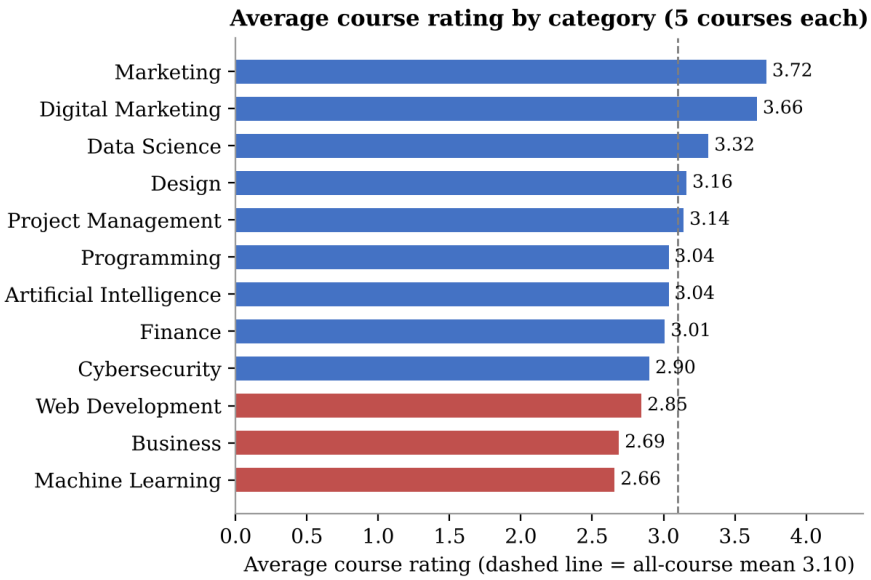


Figure 6: Average course rating by category

Table 4: Course rating by level and type

Group	Courses	Average rating	Median rating
Intermediate	18	3.32	3.28
Beginner	21	3.19	3.64
Advanced	21	2.81	2.67
Free	38	3.19	3.24
Paid	22	2.94	2.74

Interpretation. Marketing (3.72) and Digital Marketing (3.66) rate highest, and Machine Learning (2.66), Business (2.69) and Web Development (2.85) rate lowest. Advanced courses rate lowest (2.81). However, the Kruskal-Wallis tests show that the differences between categories ($p = 0.957$) and between levels ($p = 0.384$) cannot be told apart from chance, and the free-versus-paid gap is not significant either (Mann-Whitney $p = 0.514$). Price has no link with rating ($r = -0.032$).

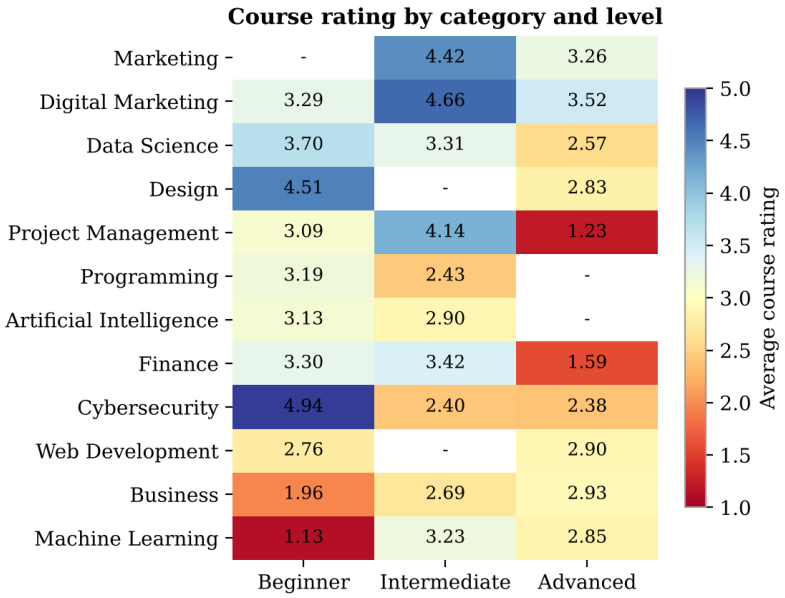


Figure 7: Course rating by category and level (heatmap)

Interpretation. The heatmap shows that quality varies sharply within a category. In Cybersecurity, the Beginner course averages 4.94 while Intermediate and Advanced courses average about 2.4. Project Management ranges from 1.23 (Advanced) to 4.14 (Intermediate), and Machine Learning from 1.13 (Beginner) to 3.23 (Intermediate). Several category-by-level cells contain a single course, so they are indicators and not firm results.

7.7 Expertise-wise instructor and course ratings

The 12 teacher expertise labels and the 12 course categories form the same set, so instructor and course ratings can be compared subject by subject.

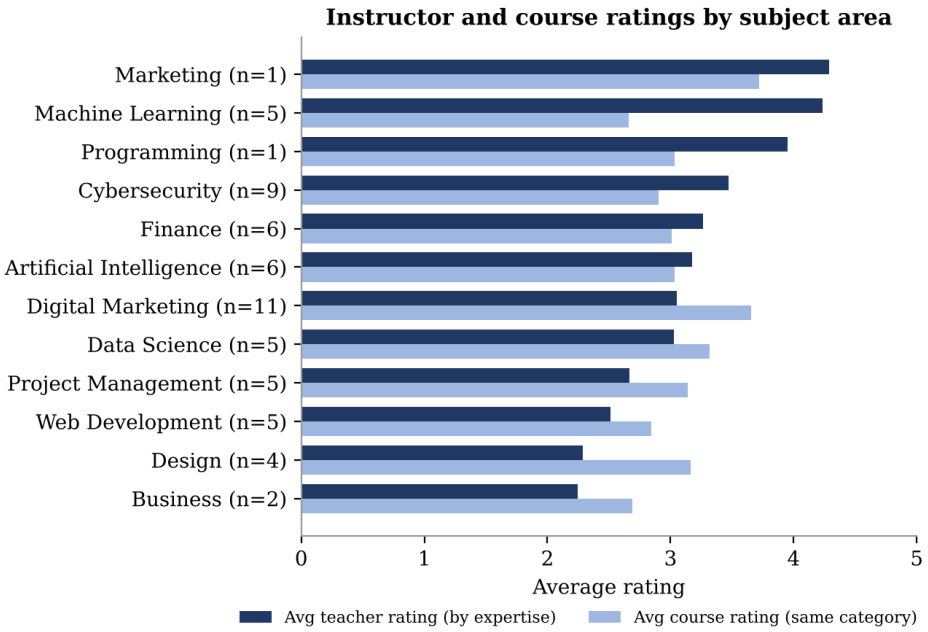


Figure 8: Instructor ratings by expertise and course ratings by category

Interpretation. Machine Learning has a high instructor rating (4.23, five teachers) but the lowest course rating (2.66). Digital Marketing shows the opposite: a middling instructor rating (3.05) with a high course rating (3.66). Business, Design and Web Development are weak on the instructor side, with averages between 2.24 and 2.51, although they rest on two to five teachers each. The Kruskal-Wallis test across expertise groups gives p = 0.067, so differences between expertise groups are suggestive and not conclusive.

7.8 Instructor performance leaderboard

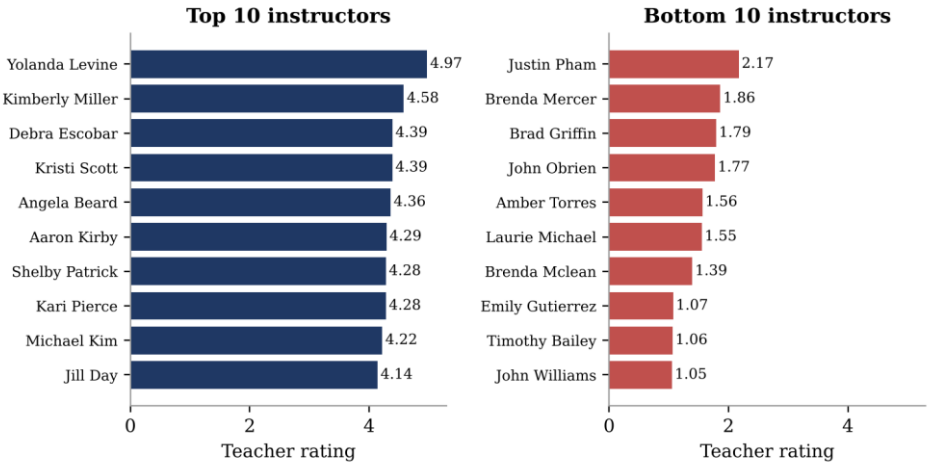


Figure 9: Top 10 and bottom 10 instructors by teacher rating

Interpretation. The ten highest-rated instructors average 11.0 years of experience, and the ten lowest-rated average 3.1 years. The leaderboard uses TeacherRating only. Transaction volume is not used as a measure of quality.

7.9 Transaction activity by instructor rating tier

Transactions per teacher range from 23 to 3,061. The average of 166.7 is misleading because two teachers, TC00042 (3,061 transactions) and TC00040 (3,025 transactions), hold 60.86% of all transactions.

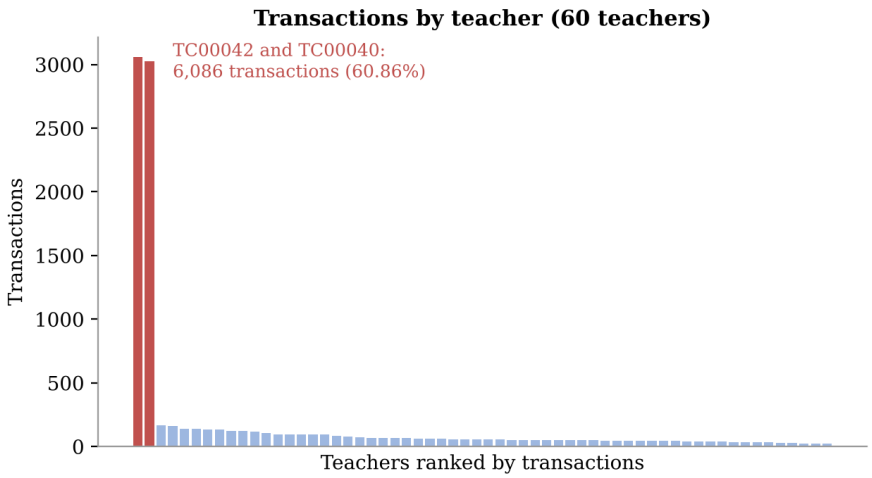


Figure 10: Transactions by teacher

These two instructors are also the two highest-rated (4.97 and 4.58), so including them creates a false impression that rating attracts volume. They share exactly the same 454 learners, as discussed in Section 10. For this reason, the relationship between rating and activity was tested both with and without them.

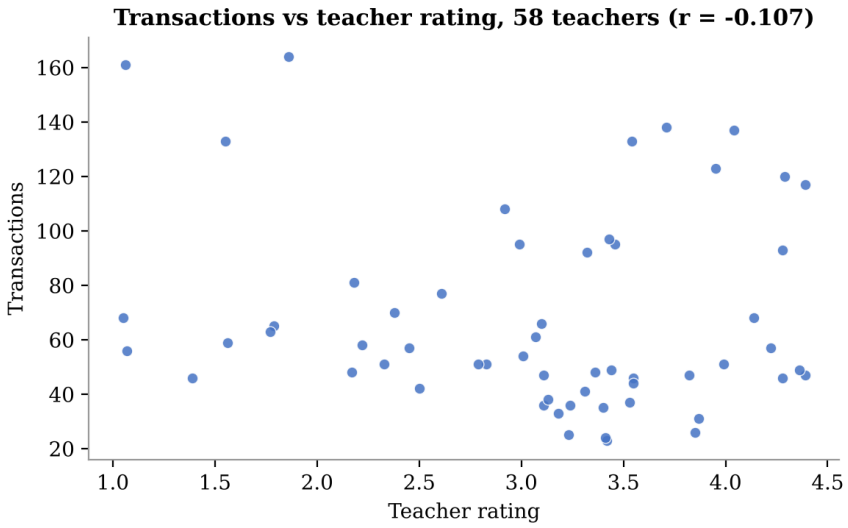


Figure 11: Transactions against teacher rating, excluding TC00040 and TC00042

Instructors were then grouped into three rating tiers: Low (below 3.0), Mid (3.0 to 3.99) and High (4.0 and above).

Table 5: Transaction activity by instructor rating tier

Tier	Teachers	Avg rating	Avg experience	Avg transactions	Median transactions
Low (<3.0)	21	2.07	2.8 yrs	76.4	63.0
Mid (3.0-3.99)	28	3.43	7.0 yrs	56.3	46.5
High (>=4.0), without TC00040 and TC00042	9	4.27	-	81.6	68.0
High (>=4.0), all 11 teachers	11	4.36	11.1 yrs	620.0	93.0

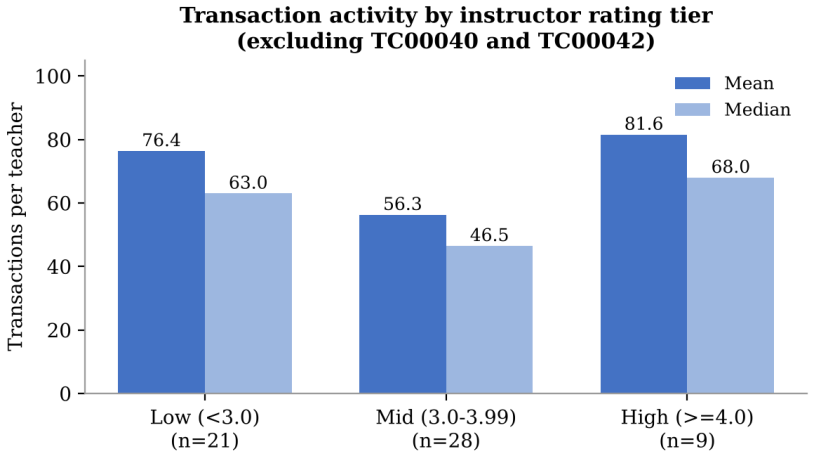


Figure 12: Transactions per teacher by rating tier, excluding TC00040 and TC00042

Interpretation. With all 11 high-rated teachers included, the high tier averages 620 transactions per teacher, but this is entirely due to the two outliers. Without them, the high tier averages

81.6 transactions, compared with 76.4 for the low tier and 56.3 for the mid tier. The mid tier has the lowest activity of all three. Activity therefore does not rise with rating. The average rating of linked courses is also almost the same across tiers (3.14 to 3.19).

8. Statistical Analysis

8.1 Descriptive statistics

Table 6: Descriptive statistics

Variable	n	Mean	Median	SD	Min	Q1	Q3	Max
Teacher rating	60	3.13	3.27	0.95	1.05	2.49	3.83	4.97
Years of experience	60	6.28	6.00	4.72	1.00	3.00	8.00	24.00
Instructor age	60	38.5	36.5	7.7	27.0	32.0	46.3	50.0
Course rating	60	3.10	3.06	1.17	1.13	2.11	4.10	4.94
Course price	60	92.99	0.00	153.60	0.00	0.00	133.62	490.90
Course duration	60	27.63	28.50	16.09	1.20	14.50	43.01	49.73

Course price has a median of zero because 38 of 60 courses are free. Course ratings are more widely spread than teacher ratings, and the middle half of course ratings lies between 2.11 and 4.10.

8.2 Correlation analysis

Table 7: Correlation results

Pair	n	Pearson r	p	Spearman ρ	p
Experience vs teacher rating	60	0.598	< 0.001	0.671	< 0.001
Age vs teacher rating	60	0.352	0.006	0.333	0.009
Experience vs linked course rating	60	-0.057	0.666	-0.048	0.715
Teacher rating vs linked course rating	60	0.000	0.997	-0.059	0.654
Transactions vs teacher rating, all teachers	60	0.318	0.013	-0.012	0.927
Transactions vs teacher rating, without TC00040 and TC00042	58	-0.107	0.425	-0.120	0.368

Direction and strength. Experience and teacher rating are positively related and the relationship is moderate ($r = 0.598$ means that about 36% of the variation in teacher rating is accounted for by a straight-line fit on experience, $R^2 = 0.358$). The result is stable: it stays at $r = 0.553$ without instructors above 14 years and at $r = 0.542$ without TC00040 and TC00042. Age is more weakly related to rating ($r = 0.352$), and because age and experience are themselves related ($r = 0.348$), the age effect is probably experience in another form. The remaining pairs are close

to zero and not significant. With all 60 teachers, transactions and rating show a weak positive Pearson value (0.318, p = 0.013), but this comes only from the two outlying teachers, and the rank-based Spearman value is near zero (-0.012).

8.3 Regression and group comparisons

A simple linear regression of teacher rating on years of experience gives a slope of 0.120 rating points per year (standard error 0.021, p < 0.001) and an intercept of 2.37. Because ratings level off after about seven years, a straight line summarises the average pattern and overstates the gain for experienced instructors.

Table 8: Group comparison tests

Comparison	Test	p-value
Course rating across 12 categories	Kruskal-Wallis	0.9572
Course rating across 3 levels	Kruskal-Wallis	0.3842
Teacher rating across 12 expertise areas	Kruskal-Wallis	0.0667
Course rating, free vs paid	Mann-Whitney U	0.5144

None of these group differences reaches the conventional 0.05 level. The expertise comparison (p = 0.067) comes closest. With only five courses per category and one to eleven teachers per expertise area, the tests have little power, so the absence of significance should not be read as proof that no differences exist.

8.4 Limits of the statistics

All relationships described in this paper are associations. A correlation between experience and rating does not show that experience causes better ratings, because subject, class size, learner allocation and rating collection could all contribute. The sample is small (60 teachers and 60 courses), p-values are shown for guidance, and the unexplained two-instructor pattern affects any measure based on transaction counts.

9. Key Performance Indicators

Five KPIs summarise instructor performance and course quality. Their formulas are chosen to be simple and transparent, and each is calculated from the 2025 data.

Table 9: Key performance indicators, formulas and results

KPI	Formula	2025 result	Interpretation
Average Teacher Rating	Sum of TeacherRating ÷ number of teachers (60)	3.125	Benchmark of teaching quality; below the 3.4 target proposed in Section 12
Average Course Rating	Sum of CourseRating ÷ number of courses (60)	3.098	Benchmark of course quality; very close to the teacher average
Rating Consistency Index (RCI)	1 − (SD of ratings ÷ 2), where 2 is the largest possible SD on a 1-5 scale	Teachers 0.525; courses 0.414	Closer to 1 means more uniform ratings. Course ratings are less consistent than teacher ratings
Experience Impact Score	Slope of the regression of TeacherRating on YearsOfExperience	0.120 points per year (r = 0.598)	Average rating gain per extra year; about 0.6 points over five years, flattening after about seven years
Enrollment Influence Ratio (transaction-activity proxy)	Average transactions per High-tier teacher (4.0 and above) ÷ average transactions per Low-tier teacher (below 3.0), without TC00040 and TC00042	1.07 (8.12 with the two outliers)	A value near 1 means that high-rated instructors record no more activity than low-rated ones

Notes on the formulas. The Rating Consistency Index is a platform-level measure of spread. It uses the sample standard deviation (0.950 for teachers and 1.171 for courses) and the maximum possible standard deviation of 2.0 for ratings between 1 and 5. A different scale would need a different denominator. The Experience Impact Score is an average slope and should be read together with the experience-band results in Section 7.4. The Enrollment Influence Ratio is labelled a proxy because a transaction is an enrolment event and not an enrolled learner: 3,000 learners made 10,000 transactions, and the 454 learners behind TC00040 and TC00042 account for most of the volume. The ratio is calculated without these two teachers for this reason.

10. Results and Findings

10.1 High- and low-rated instructors

Eleven instructors (18.3%) are rated 4.0 or above and 21 (35.0%) are rated below 3.0. Table 10 lists five from each end. The ten lowest-rated instructors all have five years of experience or less.

Table 10: Highest and lowest rated instructors (five each)

Teacher	Name	Expertise	Experience	Rating
TC00042	Yolanda Levine	Machine Learning	21 yrs	4.97
TC00040	Kimberly Miller	Cybersecurity	24 yrs	4.58
TC00004	Kristi Scott	Machine Learning	9 yrs	4.39

Teacher	Name	Expertise	Experience	Rating
TC00037	Debra Escobar	Finance	8 yrs	4.39
TC00007	Angela Beard	Machine Learning	6 yrs	4.36
TC00043	John Williams	Project Management	1 yr	1.05
TC00028	Timothy Bailey	Business	3 yrs	1.06
TC00045	Emily Gutierrez	Web Development	5 yrs	1.07
TC00036	Brenda Mclean	Digital Marketing	4 yrs	1.39
TC00027	Laurie Michael	Artificial Intelligence	3 yrs	1.55

Table 11: Low ratings by experience band

Experience	Teachers	Rated below 3.0	Share	Lowest rating
1 to 3 yrs	19	13	68.4%	1.05
4 to 6 yrs	17	8	47.1%	1.07
7 to 9 yrs	14	0	0.0%	3.07
10+ yrs	10	0	0.0%	3.23

All 21 instructors below 3.0 have six years of experience or less, and none of the 24 instructors with seven or more years is below 3.0. Rating risk therefore sits with newer instructors: 21 of the 36 instructors with six years or less (58.3%) are rated below 3.0.

Instructor gender shows a small difference. The 27 female instructors average 3.23 and the 33 male instructors average 3.04, and the female group also has more average experience (6.96 against 5.73 years). No significance test was run, and the difference in experience could explain part of the gap. When transactions are used to link instructor gender to course level, the average rating of linked courses is similar for both groups (female 2.80 Advanced, 3.10 Beginner, 3.33 Intermediate; male 2.84, 3.37 and 3.34), and these figures are influenced by the two high-volume instructors.

10.2 Top- and low-performing course categories

Marketing (3.72) and Digital Marketing (3.66) are the highest-rated categories. Machine Learning (2.66), Business (2.69) and Web Development (2.85) are the lowest. Because the category differences are not statistically significant, the more useful finding is the spread within categories. Weak courses appear in 11 of the 12 categories, and the best and worst course in the same category can be 3.5 to 3.6 points apart (Project Management 1.23 to 4.84, Machine Learning 1.13 to 4.73, Cybersecurity 1.35 to 4.94, Programming 1.28 to 4.74). Quality is a course-level issue, and reviews should target individual courses.

Thirteen courses are rated below 2.0 and twenty are rated below 2.5, which is one third of all courses. Table 12 lists the 13 priority courses.

Table 12: Courses rated below 2.0

Course	Name	Category	Level	Type	Rating
CR00029	Natural Language Processing	Machine Learning	Beginner	Free	1.13
CR00046	AI for Beginners	Artificial Intelligence	Beginner	Paid	1.20
CR00039	Project Planning	Project Management	Advanced	Paid	1.23
CR00005	Full Stack Development	Programming	Beginner	Free	1.28
CR00051	HTML & CSS	Web Development	Advanced	Paid	1.30
CR00034	Data Encryption	Cybersecurity	Intermediate	Paid	1.35
CR00040	Leadership in Projects	Project Management	Beginner	Paid	1.51
CR00008	Photoshop Mastery	Design	Advanced	Free	1.52
CR00045	Financial Modeling	Finance	Advanced	Free	1.59
CR00030	Reinforcement Learning	Machine Learning	Intermediate	Free	1.73
CR00027	Neural Networks	Machine Learning	Advanced	Free	1.81
CR00031	Network Security	Cybersecurity	Intermediate	Free	1.91
CR00015	Leadership Essentials	Business	Beginner	Paid	1.96

Six of these courses are paid. Together they carry 2,102 transactions (21.0%) and 263,211 in revenue (28.9%).

10.3 Experience and expertise patterns

Experience is the clearest pattern in the data (Section 7.4). Expertise shows a weaker one. Machine Learning (4.23), Marketing (4.29, one teacher) and Programming (3.95, one teacher) have the highest average instructor ratings, and Business (2.24, two teachers), Design (2.29), Web Development (2.51) and Project Management (2.66) have the lowest. Web Development, one of the lowest-rated groups, also has the least average experience (4.0 years), which suggests that experience and subject are partly entangled.

Teaching assignments mostly follow expertise. A transaction is "matched" when the instructor's expertise equals the category of the course. Overall, 41.11% of transactions are matched, well above the 8.34% expected by chance. The other 58 instructors match 80.43% of the time, while TC00040 and TC00042 match only about 16% because they appear with 55 of the 60 courses each. These two instructors account for 87.0% of all mismatched transactions. Three Digital Marketing instructors (TC00050, TC00013 and TC00020) match under 50%. Supply is also uneven: Digital Marketing has 11 instructors for five courses, while Marketing and Programming have one instructor each.

10.4 Comparison of instructor rating tiers

Table 5 and Figure 12 compare the tiers. The tiers differ clearly in rating and experience, since the High tier averages 11.1 years against 2.8 years for the Low tier. They do not differ in the rating of linked courses (3.14 to 3.19) or in transaction activity once the two outliers are set aside. Learner reach of weak instructors is also wide: about 1,098 learners (36.6% of all learners) were linked at least once to an instructor rated below 2.5, although those instructors hold only 11.8% of transactions.

10.5 Revenue and exposure to weak courses

Revenue comes only from paid transactions: 3,597 transactions on 22 paid courses by 1,605 learners, a total of 911,323.47 in the dataset's own currency units. Artificial Intelligence (22.2%), Business (19.9%) and Project Management (18.6%) together give 60.7% of revenue, and three courses (Computer Vision, Business Strategy and Python Basics) give 27.1%. Machine Learning earns almost nothing (133.38) because its only paid course is priced at 0.78.

Table 13: Transactions and revenue by course rating band

Course rating	Courses	Transactions	Share	Revenue	Share
Below 2.0	13	2,102	21.0%	263,211	28.9%
2.0 to 2.99	16	2,602	26.0%	154,856	17.0%
3.0 to 3.99	15	2,573	25.7%	266,050	29.2%
4.0 and above	16	2,723	27.2%	227,206	24.9%

Weak courses are not marginal. The 20 courses rated below 2.5 (seven of them paid) bring 29.9% of revenue and 32.3% of transactions, and 1,520 learners (50.7%) took at least one of them. Twelve of the 22 paid courses are rated below the all-course mean and bring 45.9% of revenue.

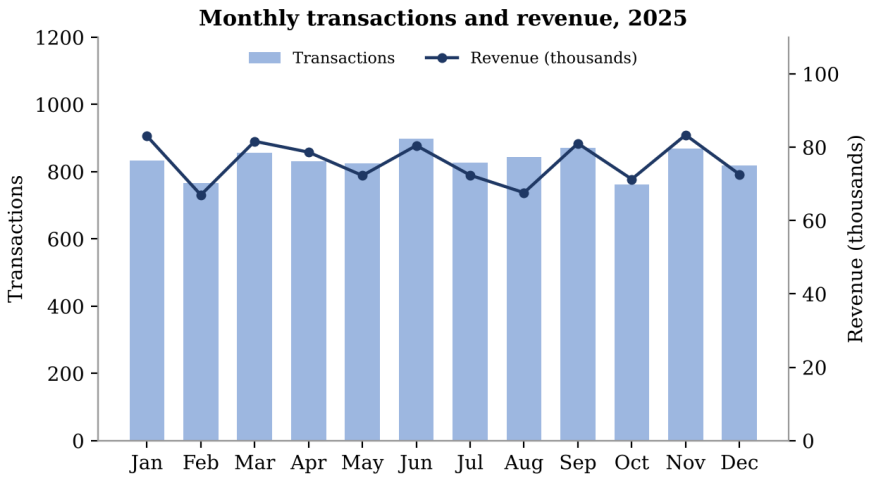


Figure 13: Monthly transactions and revenue, 2025

Demand is flat across 2025. Monthly transactions range from 762 in October to 899 in June, a variation of 4.8% around the mean, and the first and second halves of the year are almost equal (5,011 and 4,989 transactions). With one year of data, seasonality cannot be judged.

10.6 Data-quality concerns

The most important concern is the pattern in TC00040 and TC00042. Together they hold 60.86% of transactions and 62.18% of revenue, yet they share exactly the same 454 learners (15.1% of all users). These learners average 13.41 transactions each, against 1.54 for the other 2,546 learners. Each of the two instructors appears with 55 of the 60 courses and skips exactly one category: TC00040 has no Machine Learning transactions and TC00042 has no Cybersecurity transactions. The cause is not known. It may be a real cohort, a default instructor assignment or a feature of how the data was recorded. It should be confirmed with the data owner before it drives any decision. The data was kept as it is, and results are shown with and without these two instructors wherever it matters.

Other concerns are gaps in what the data records: no TeacherID on Courses, no review count behind each rating, no dates on ratings, no completion, outcome or refund data, no stated currency or duration unit, one year of data only, and an unusually low price for course CR00028.

11. Discussion

11.1 Interpretation in the context of online education

The findings show that EduPro's teaching staff is uneven. Instructors with limited experience carry most of the rating risk, which fits the intuition that teaching skill grows with practice.

The levelling-off after about seven years suggests that the early years matter most, and that structured support for new instructors could shorten the learning curve. A more surprising finding is that instructor rating and course rating are unrelated in this data. In online learning the course is a designed product, with its own content, sequence and level, and a strong teacher cannot rescue a poorly designed course, nor can a weaker teacher always spoil a well-designed one. The wide spread of course ratings inside every category supports this view, although the weak link between teachers and courses in the data means it cannot be proven here.

The transaction findings point to a discovery problem. Among 58 of the 60 instructors, learners enrol with low-rated and high-rated instructors at similar rates, so ratings do not appear to guide learner choice. If ratings were shown clearly and used in recommendations, one would expect highly rated instructors to attract more activity. This is an explanation that the data cannot confirm, and it is listed as a hypothesis to test.

11.2 Practical implications for EduPro

- Quality problems reach many learners and much revenue: half of the learners took a course rated below 2.5, and 29.9% of revenue comes from such courses. Quality reviews therefore have direct commercial value.
- Dependence on two instructors, if real, is a business risk. If it is a recording artefact, then transaction-based reports are distorted until it is fixed.
- Instructor evaluation should not use transaction volume. It does not follow rating, and it is dominated by two outliers.
- Course review should be done course by course. Category-level rules would miss the large differences inside categories.

11.3 Correlation and causation

Every relationship in this paper is an association. The link between experience and rating could reflect teaching skill gained over time, but it could equally reflect which instructors remain on the platform, how learners are allocated to newer and older instructors, or how ratings are collected. The statistics cannot separate these explanations. For this reason, experience is best used as a signal for where to offer support, and not as a hiring or promotion rule.

11.4 Sample size and dataset limitations

With 60 instructors, 60 courses, five courses per category and one to eleven instructors per expertise area, small groups can be moved by a single record, and statistical tests have limited

power. The absence of significant differences between categories and levels should not be read as proof that none exist. In addition, the missing instructor link on courses, the absence of review counts and dates, and a single year of data restrict what can be concluded about causes and trends.

12. Recommendations

The recommendations are proposals for management. They are grouped under six areas of action, and each is linked to a KPI so that progress can be measured. Table 14 gives the priorities and timeframes.

12.1 Instructor training and mentoring

Run a development programme for early-career instructors, starting with the nine instructors rated below 2.0 and then the 21 rated below 3.0. Pair instructors with fewer than seven years of experience with mentors, review their lowest-rated courses first, and test whether the programme closes the gap before experience is used for any hiring decision (R4).

12.2 Course-content improvement

Introduce a course quality review process. Start with the 13 courses rated below 2.0, paid courses first because learners paid for them, then the seven on the watch list between 2.0 and 2.49. Gate the promotion of paid courses rated below 2.5. Review content, length, level fit and prerequisites, regardless of who teaches the course (R3).

12.3 Targeted support for low-rated categories

Give extra attention to Machine Learning (2.66), Business (2.69) and Web Development (2.85), and to Advanced courses (2.81). Because ratings vary widely inside every category, support should be directed to the specific weak courses inside them. Confirm the price of CR00028 and set an expertise-alignment policy for assignments, since instructors working outside their field account for most mismatches (R3, R6).

12.4 Recognition of high-performing instructors

Recognise and retain the 11 instructors rated 4.0 or above, and use them as peer mentors for early-career instructors. Place them more visibly in search and recommendations, and use ratings in course placement, so that ratings begin to guide learner choice (R4, R7).

12.5 Regular monitoring through KPIs

Adopt an evaluation scorecard that excludes transaction volume and refresh the KPI pack monthly and quarterly, using the dashboard (R5, R8). Table 15 gives a selection of the KPIs with 2025 baselines and proposed targets. The targets are proposals and not existing commitments.

12.6 Improved data collection and quality assurance

Validate the two-instructor pattern with the data owner within 30 days (R1). Add a TeacherID to the Courses sheet, store the number of reviews behind each rating, keep dated rating snapshots, and record completion, outcome and refund data, currency and duration units (R2). These additions would remove most of the limitations of this study.

Table 14: Recommendations, priorities and timeframes

ID	Recommendation	Priority	Timeframe	Owner
R1	Validate the two-instructor transaction pattern	P1	0-30 days	Data owner, Operations
R2	Close the data gaps: teacher-course link, review counts, dated ratings	P1	0-90 days	Data owner
R3	Introduce a course quality review process; gate promotion of paid courses below 2.5	P1	0-90 days	Content quality lead
R4	Run a development and mentoring programme for early-career instructors	P2	30-180 days	Learning and development
R5	Adopt an evaluation scorecard that excludes transaction volume	P2	30-120 days	Academic quality lead
R6	Set an expertise-alignment policy for assignments	P2	60-180 days	Operations
R7	Show ratings in learner choice and test whether activity follows	P3	90-270 days	Marketing, Product
R8	Start monthly and quarterly KPI monitoring	P2	60-120 days	Data / BI analyst

Table 15: Selected KPIs with 2025 baseline and proposed target

ID	KPI	Baseline	Proposed target	Horizon
K01	Average teacher rating	3.125	3.4 or more	12 months
K02	Share of teachers rated below 3.0	35.0%	25% or less	12 months
K03	Teachers rated below 2.0	9	3 or fewer	6 months
K05	Share of teachers rated 4.0 or above	18.3%	25% or more	12 months
K07	Average course rating	3.10	3.4 or more	12 months
K08	Courses rated below 2.0	13	6 or fewer	12 months
K09	Share of courses rated below 2.5	33.3%	20% or less	12 months
K10	Revenue share from paid courses below 2.5	29.9%	15% or less	12 months
K14	Teachers matching under 70% of transactions	24	12 or fewer	12 months

ID	KPI	Baseline	Proposed target	Horizon
K16	Rating-volume rank correlation (58 teachers)	-0.12	0.3 or more	12 months

Three indicators are for monitoring only and have no numeric target: the experience-rating association ($\rho = 0.671$), the share of transactions held by the top two instructors (60.86%) and the share of learners linked to them (15.1%). Targets for the last two should be set only after the pattern is validated.

13. Streamlit Dashboard Description

The dashboard lets non-technical stakeholders explore the same data. It was built with Streamlit, pandas and Plotly and deployed on Streamlit Community Cloud. It loads the EduPro workbook automatically, so no upload is needed.

Live dashboard: https://edupro-quality-dashboard-kani-2026-3yzchnyaqxhaqhmwdgsucq.streamlit.app/

13.1 Modules

- Executive Overview: counts of instructors, courses and transactions, average teacher and course rating, average experience, the free-versus-paid split and the monthly transaction trend.
- Instructor Performance: instructor performance leaderboard (top 10 and bottom 10 by teacher rating), teacher rating distribution, transaction volume by instructor, and a scatter plot of experience against teacher rating coloured by expertise.
- Course Quality: average course rating by category and by level, rating distribution by course type, and tables of the highest- and lowest-rated courses.
- Expertise Analysis: instructor distribution by expertise with average rating and experience, and the exact-label match between instructor expertise and course category.
- Enrollment and Transactions: recorded amounts, paid and free activity, transactions by category and level, and each instructor's share of transactions.

13.2 Filters and controls

- Instructor expertise and instructor gender selectors.
- Course category and course level selectors.
- A teacher rating range slider and a transaction date range selector.

- CSV download of the filtered tables.

13.3 Data-quality checks

A "Data validation and metric definitions" panel reports the sheets loaded, the number of transaction rows, unmatched course and instructor references, duplicate transaction IDs and invalid dates, and warns the user if any of these are found. Short notes beneath the charts state what each chart shows and what it does not show. For example, the leaderboard note states that transaction volume must not be read as teaching quality, and the expertise note states that a label match is not a validated skill match.

The category-by-level heatmap in this paper (Figure 7) is not part of the current dashboard and could be added as a future improvement.

13.4 Dashboard screenshots

Figures 14 to 16 show the deployed dashboard as captured on 22 October 2026. The screenshots were taken with sidebar filters applied, so the numbers on the cards describe the selected group of instructors and courses and not the whole platform. The platform-wide results are those given in Sections 7 to 10.

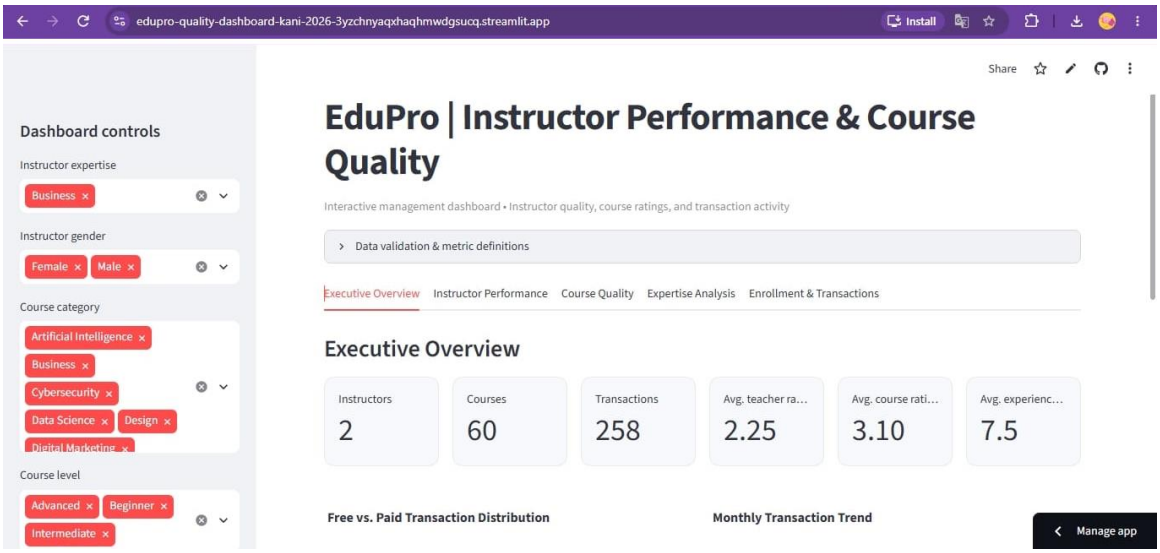


Figure 14: Executive Overview with the Business expertise filter applied

Interpretation. With the instructor expertise filter set to Business, the overview cards show 2 instructors, 60 courses, 258 transactions, an average teacher rating of 2.25, an average course rating of 3.10 and an average experience of 7.5 years. This is consistent with Section 10.3, where Business is one of the lowest-rated expertise groups and has two instructors.

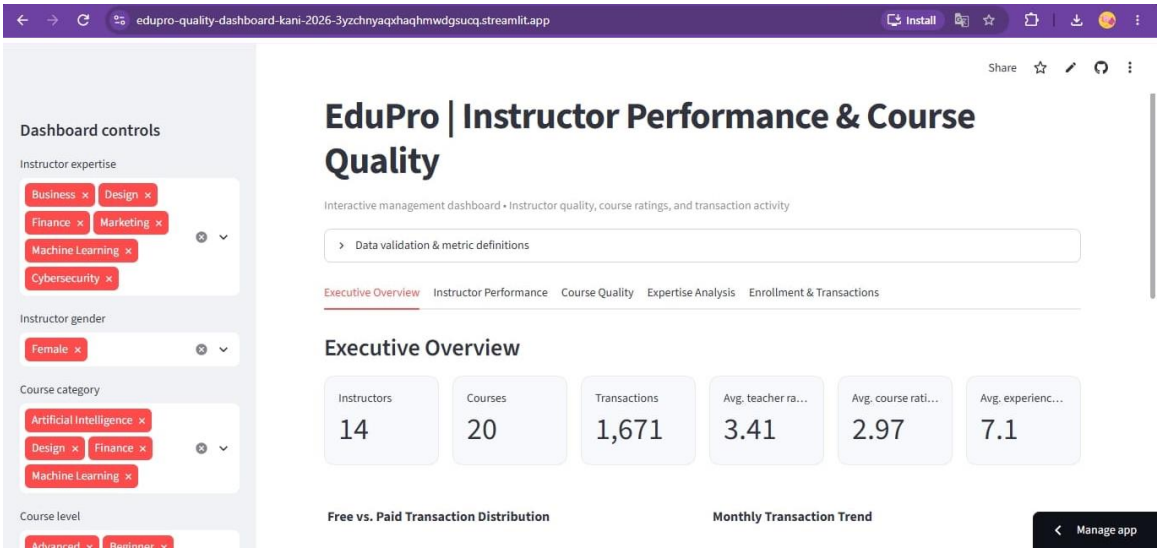


Figure 15: Executive Overview with the Female instructor filter, six expertise areas and four course categories selected

Interpretation. With female instructors, six expertise areas (Business, Design, Finance, Marketing, Machine Learning and Cybersecurity) and four course categories (Artificial Intelligence, Design, Finance and Machine Learning) selected, the cards show 14 instructors, 20 courses, 1,671 transactions, an average teacher rating of 3.41, an average course rating of 2.97 and an average experience of 7.1 years. The sidebar controls can be combined in this way to compare any group of instructors and courses.

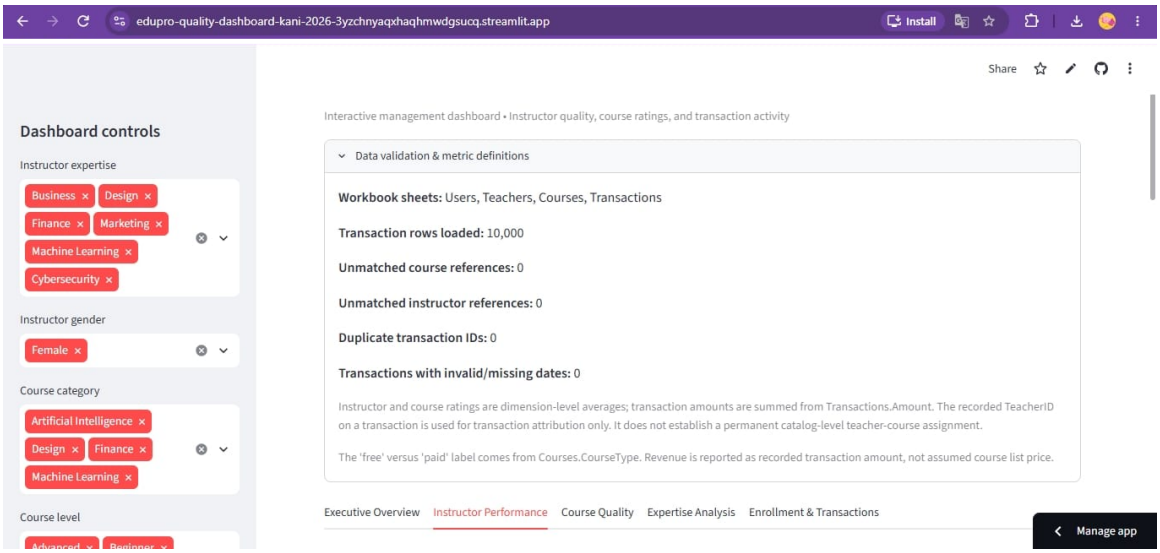


Figure 16: Data validation and metric definitions panel

Interpretation. The panel confirms that the four workbook sheets (Users, Teachers, Courses and Transactions) were loaded with 10,000 transaction rows. It reports 0 unmatched course references, 0 unmatched instructor references, 0 duplicate transaction IDs and 0 transactions with invalid or missing dates, which agrees with Table 2. The notes beneath the checks state

that ratings are dimension-level averages and that amounts are summed from Transactions.Amount. They also state that the TeacherID on a transaction is used for attribution only and does not establish a permanent teacher-course assignment, and that the free or paid label comes from Courses.CourseType, with revenue reported as the recorded transaction amount and not the course list price. These notes match the limits described in Section 5.2.

14. Conclusion

This study built a data-driven picture of instructor performance and course quality on EduPro. Teaching quality is uneven: only 18.3% of instructors reach a rating of 4.0, while 35.0% fall below 3.0, and all of the lower group have six years of experience or less. Experience has a moderate positive association with rating, equal to about 0.12 points per year and levelling off after about seven years. Course quality is a separate matter. Instructor rating does not predict the rating of the courses linked to the instructor, and course quality varies far more between individual courses than between categories. The weakest courses reach half of the learners and bring in nearly a third of revenue. Learners are not being guided towards better teaching, since transaction activity does not follow rating among 58 of the 60 instructors.

Two things limit how far these results can go. The platform cannot link a course to its instructor, and two instructors share one group of 454 learners for reasons the data does not explain. For this reason, the first actions recommended are to validate that pattern and to close the data gaps, followed by a course-by-course quality review and support for early-career instructors. With those foundations in place, the KPI framework and the dashboard give EduPro a practical way to move instructor evaluation from impressions to evidence.

15. Limitations and Future Scope

15.1 Limitations

- There is no TeacherID on Courses, so course quality cannot be attributed to an instructor, and any statement such as "this instructor's course is weak" cannot be supported.
- Ratings are single values per instructor and per course, with no review counts and no dates. Reliability and change over time cannot be tested.
- There is no completion, outcome, refund or retention data, so the study cannot say whether low ratings cost revenue or learners.

- The groups are small, and statistical tests have limited power. Differences between categories or expertise groups are indicative.
- Two instructors share the same learners, and this may be a recording artefact. Transaction-based measures are affected by it.
- Only 2025 is covered, so year-on-year change and seasonality cannot be judged.
- The rating scale is assumed to run from 1 to 5, and currency and duration units are not stated.
- All relationships are associations. No causal claim is made.

15.2 Future scope

- Add a teacher-course assignment table so that course quality can be attributed to instructors.
- Collect review counts and dated rating snapshots, and then weight ratings by reliability and track improvement over time.
- Add completion, assessment results and refund data, to measure learning quality and not only enrolment.
- Analyse review text to find the reasons behind low ratings.
- Extend the data to several years to test trends and seasonality, and then build predictive models of course rating.
- Add the category-by-level heatmap and alerts for instructors and courses that fall below a set rating to the dashboard.

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17. Appendix

Appendix A: Data dictionary

The full data dictionary is given in Table 1 (Section 5.1).

Appendix B: KPI formulas

Table 16: KPI formulas

KPI	Formula
Average Teacher Rating	mean(TeacherRating) over the 60 instructors
Average Course Rating	mean(CourseRating) over the 60 courses
Rating Consistency Index	RCI = 1 − SD(rating) ÷ SD_max, with SD_max = (5 − 1) ÷ 2 = 2. Teachers: 1 − 0.950 ÷ 2 = 0.525. Courses: 1 − 1.171 ÷ 2 = 0.414
Experience Impact Score	b1 in TeacherRating = b0 + b1 × YearsOfExperience (ordinary least squares); b1 = 0.120, b0 = 2.368, R^2 = 0.358
Enrollment Influence Ratio (proxy)	EIR = mean transactions per teacher with TeacherRating ≥ 4.0 ÷ mean transactions per teacher with TeacherRating < 3.0, without TC00040 and TC00042. EIR = 81.6 ÷ 76.4 = 1.07
Expertise match rate	Transactions where Teacher.Expertise = Course.CourseCategory ÷ all transactions

Appendix C: Python libraries used

- pandas: loading the workbook, joins, grouping and summary tables.
- NumPy: numerical operations and the linear fit.
- SciPy: Pearson and Spearman correlation, linear regression, Kruskal-Wallis and Mann-Whitney tests.
- Matplotlib: the charts in this paper.
- Plotly: interactive charts in the dashboard.
- Streamlit: the dashboard application.
- openpyxl: reading the Excel workbook.

Appendix D: Important data-cleaning and analysis steps

- 1. Load the four sheets and parse dates and numeric fields.
- 2. Check missing values, duplicate identifiers, duplicated rows and unmatched keys on every sheet.
- 3. Join Transactions to Courses (CourseID) and Teachers (TeacherID) with many-to-one checks, and confirm that 10,000 rows remain.

- 4. Build teacher-level tables: rating, experience, transactions, unique learners and average rating of linked courses.
- 5. Calculate descriptive statistics, correlations, the regression and group tests.
- 6. Run the correlations again without TC00040 and TC00042, and without instructors above 14 years, to test stability.
- 7. Group instructors into rating tiers and compare transaction activity.
- 8. Calculate the five KPIs and produce the charts.

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